

# Aerodynamics, Turbulence, and Computational Fluid Dynamics

A gentle, first-year-friendly introduction

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## How to Use This Handout

This version keeps the math to basic algebra, ratios, and simple single-variable rates of change — nothing beyond what a first-year calculus course covers. The goal is to build the physical picture first. A companion version of this handout exists with the full multivariable-calculus treatment (vector calculus, partial differential equations) for when you’re ready for it; everything here is a true simplification, not a different story.

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# 1 The Fluid as a Continuum

Air is made of individual molecules, but there are about  $10^{25}$  of them in a single cubic meter — far too many to track one by one. So instead of following molecules, we describe air using smooth, averaged quantities that vary from point to point: density, pressure, velocity, and temperature. At any location and any moment you can ask "what is the density here, right now?" and get a sensible answer, the same way you can ask "what is the local population density of this city block?" without listing every resident by name.

This is called treating the fluid as a continuum, and it works extremely well for anything an aircraft, car, or building experiences. It only breaks down in extreme situations, like air so thin (high in the atmosphere) that individual molecules travel long distances between collisions.

Fluid mechanics replaces billions of individual molecules with a handful of smooth, averaged quantities that vary in space and time.

## 1.1 Two Ways to Watch a Flow

There are two natural ways to describe a moving fluid. You could stand at a fixed spot and record what the flow does as it passes you — like standing on a bridge and watching the river go by beneath you. Or you could follow one specific bit of fluid on its journey — like dropping a leaf in the river and following it downstream.

Both describe the same physics, just organized differently. CFD software almost always uses the first approach (fixed viewing points), because it's much easier to store numbers on a fixed grid than to track billions of individually moving fluid parcels.

# 2 Conservation of Mass

Draw an imaginary box somewhere in the flow. Mass can't be created or destroyed, so if the amount of mass inside the box is staying the same, then whatever flows in through one side must flow out through another. If more flows in than out, the mass (and therefore the density) inside the box has to increase.

For a steady flow through a pipe or duct where the density doesn't change, this gives a simple and very useful rule: the volume flow rate is the same everywhere along the pipe,

$$A_1 V_1 = A_2 V_2,$$

where  $A$  is the cross-sectional area and  $V$  is the average speed. If the pipe narrows,  $A$  gets smaller, so  $V$  must get larger to keep the product the same. This is exactly why water speeds up when you put your thumb over the end of a hose, and it's the same reason air speeds up as it moves through a converging nozzle.

For a fluid like air, where density stays essentially constant at low speeds, this idea — what goes in must come out — is really the entire content of "conservation of mass." At higher (compressible) speeds density can change too, but the bookkeeping principle is identical: track everything crossing the boundary of your box.

### 3 Conservation of Momentum: Newton’s Second Law for Fluids

Newton’s second law,  $F = ma$ , applies to any chunk of matter, and a small blob of fluid is no exception. The forces acting on a fluid blob are:

- **Pressure forces:** fluid gets pushed from regions of high pressure toward regions of low pressure, the same way a ball rolls downhill from high to low elevation.
- **Viscous forces:** neighbouring layers of fluid moving at different speeds drag on each other through friction, the faster layer tending to speed up the slower one and vice versa.
- **Gravity:** usually small for air compared to the other two forces, except when dealing with buoyancy-driven flows.

Adding these up and setting the result equal to mass times acceleration is, in essence, the entire content of the (much more intimidating-looking) Navier–Stokes equations you’ll meet in later courses. Nothing conceptually new is happening — it’s just Newton’s second law, applied to something squishy and continuously deforming instead of a rigid block.

#### 3.1 A Subtlety: A Fluid Can Accelerate Even in “Steady” Flow

This trips up almost everyone the first time they see it. “Steady flow” means the pattern of the flow — the picture you’d see if you took a long-exposure photo — isn’t changing with time. It does *not* mean nothing is accelerating.

Picture steady flow through a garden-hose nozzle that narrows toward the tip. The overall picture of the flow never changes over time (steady), yet any individual bit of water clearly speeds up as it approaches the narrow tip, because it’s moving into a region where the flow is simply faster. That’s real acceleration, caused entirely by the water changing location, not by the flow pattern changing over time. Fluids can accelerate by moving into a faster region of an otherwise unchanging flow field — this is called convective acceleration, and it’s responsible for a great deal of aerodynamic behaviour, including the low pressure over the top of a wing (via Bernoulli’s equation, below).

#### 3.2 Viscosity, in Plain Terms

For a simple case — fluid sliding in flat layers, one above another — the friction-like stress between two layers is

$$\tau = \mu \frac{\Delta u}{\Delta y},$$

where  $\Delta u$  is the difference in speed between two nearby layers separated by a small distance  $\Delta y$ , and  $\mu$  is the fluid’s viscosity, a property that tells you how “sticky” or resistant to shearing the fluid is (honey has a large  $\mu$ ; air has a very small one). The bigger the speed difference between neighbouring layers, and the closer together they are, the larger the friction-like stress between them.

## 4 Reynolds Number: Inertia vs. Friction

Take a characteristic size  $L$  for the object in the flow (a wing chord, a pipe diameter) and a characteristic speed  $U$ . The Reynolds number is the ratio

$$Re = \frac{\rho U L}{\mu}.$$

You can think of the numerator as roughly measuring how strongly the flow's own momentum (inertia) wants to keep it moving, and the denominator as measuring how strongly friction (viscosity) wants to damp that motion out. A large  $Re$  means inertia wins — the flow coasts along on its own momentum, the way a bowling ball glides across ice. A small  $Re$  means friction wins — the flow is heavily damped, the way a marble creeps through honey.

This single number is why intuition built from watching thick, slow fluids doesn't transfer directly to air moving over a wing: a honeybee's wing and a cruising submarine both obey the same physical laws, but sit at wildly different Reynolds numbers, so they "feel" very different kinds of flow. High- $Re$  flows are also the ones prone to the kind of instabilities that eventually turn into turbulence, covered later on.

## 5 Where Lift and Drag Come From

Every aerodynamic force ultimately comes from just two things acting on a surface: pressure, pushing perpendicular to it, and friction (viscous shear), dragging along it. Lift is mostly a pressure effect — lower pressure above a wing, higher pressure below. Drag has both a pressure part (from the wake behind an object) and a friction part (from air rubbing along the surface).

Rather than working with raw forces, which depend on the size of the object and how fast it's going, we usually work with dimensionless coefficients:

$$L = \frac{1}{2}\rho V^2 S C_L \quad \text{and} \quad D = \frac{1}{2}\rho V^2 S C_D.$$

The term  $\frac{1}{2}\rho V^2$  is called dynamic pressure — it's basically the kinetic energy packed into each unit volume of the oncoming air — and  $S$  is a reference area, so multiplying the two gives units of force. All the messy details about shape, angle, and flow condition then get bundled into the single numbers  $C_L$  and  $C_D$ . This is exactly why wind-tunnel testing works: you can test a small model, measure  $C_L$  and  $C_D$ , and use those same numbers to predict the forces on the full-size vehicle.

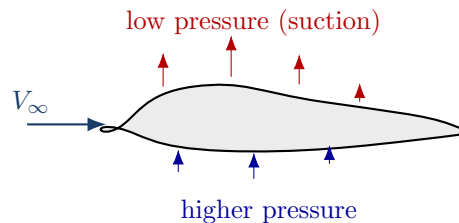


Figure 1: Pressure distribution around a lifting airfoil. Lower pressure above and higher pressure below combine into a net upward force.

## 6 Lift as a Push on the Air

There's a second, equally valid way to think about lift: instead of looking at the pressure on the wing's surface, look at what the wing does to the air around it. Air approaching the wing is deflected slightly downward as it passes; the wing has pushed the air down, and so, by Newton's third law, the air pushes the wing up by exactly the same amount. This is precisely the same idea as a garden hose recoiling in your hand when you turn a water jet: redirecting momentum requires (and produces) a force.

These are two different bookkeeping methods — adding up pressure over the wing's surface, versus tracking the momentum given to the surrounding air — but they always give exactly the same answer, because both are really just Newton's laws applied to the same physical situation.

## 7 Circulation and the Kutta–Joukowski Idea

Imagine drawing a big loop in the air around an airfoil and, at every point along the loop, recording how much the local flow is pushing "with" your direction of travel versus "against" it, then adding everything up. If the air has a net swirling tendency in the same sense as your loop, you get a large positive total; if not, the contributions mostly cancel. This total is called circulation, usually written  $\Gamma$ .

An airfoil with a sharp trailing edge, tilted at some angle to the oncoming flow, forces the air to leave smoothly off the back edge rather than curling around it. It turns out that satisfying this condition (called the Kutta condition) requires the flow around the airfoil to carry a specific amount of circulation. There's a remarkably clean formula connecting that circulation directly to lift:

$$L' = \rho V_\infty \Gamma.$$

More circulation means more lift. This gives a third way to think about lift — alongside pressure and momentum — and for idealized, simplified airfoil shapes it can actually be calculated exactly, using the mapping trick described next.

## 8 Vorticity: Local Spin in the Flow

Vorticity describes how much a tiny patch of fluid is locally spinning, at a single point — imagine dropping a miniature paddle wheel into the flow at that point and watching how fast it spins. This is different from circulation, which is a property of a whole loop, not a single point. A flow can have strong local spin (vorticity) tightly concentrated in a thin region — like right next to a wing's surface, or in the tight core trailing off a wingtip — while being essentially non-spinning everywhere else. The loop integral defining circulation still comes out large as long as the loop encloses that spinning region, even though most of the loop itself passes through non-spinning air.

## 9 Ideal (Frictionless) Flow and the Joukowski Trick

If we temporarily ignore both viscosity and any local spin in the flow, the mathematics simplifies enormously, to the point where entire families of problems can be solved with pencil and paper using tools from complex numbers. One of the cleanest results is the Joukowski transformation,

$$z = \zeta + \frac{a^2}{\zeta},$$

a formula that takes points in one plane ( $\zeta$ ) and maps them to points in another plane ( $z$ ). Remarkably, a plain circle in the  $\zeta$  plane gets stretched by this formula into a shape that looks just like a cambered airfoil, complete with a sharp trailing edge, in the  $z$  plane.

Since the frictionless flow around a simple circle has been solvable since the 1800s, this trick lets you solve the easy circle problem and then carry that solution along through the transformation to get an exact answer for the (much more complicated-looking) airfoil, without ever solving a hard problem directly. The price of this simplicity is that ignoring viscosity means the theory can't predict drag or stall — those effects come specifically from viscosity, which this idealized picture leaves out.

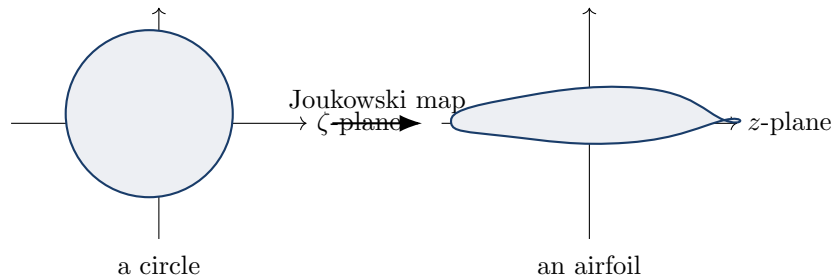


Figure 2: The Joukowski transformation stretches a circle into an airfoil shape, carrying the easy circle solution along with it.

## 10 No-Slip: Why Viscosity Matters Near a Wall

Real air has some viscosity, however small, and right at a solid surface, the air molecules essentially stick to it: the air's speed relative to the wall is exactly zero. This is called the no-slip condition. A short distance away, the air might be moving at hundreds of kilometres per hour, so the speed has to climb from zero at the wall up to the free-stream value over some small distance.

Because that change in speed happens over a very short distance, the speed *gradient* near the wall — how quickly speed changes with distance from the wall — can be very large, even for a low-viscosity fluid like air. And since the friction-like stress depends on that gradient (recall  $\tau = \mu \Delta u / \Delta y$ ), a thin region near the wall ends up mattering a great deal, even though viscosity itself is small. That thin region is the boundary layer.

## 11 Boundary Layers

At high Reynolds number, it turns out that viscosity's effects are mostly confined to a thin layer hugging the surface; everywhere outside that layer, the flow behaves almost as if it had no viscosity at all. This was a huge insight (due to Ludwig Prandtl, in 1904) because it means you can treat the problem as two much simpler pieces: an outer region where you can use the easy, frictionless equations, and a thin inner region near the wall where viscosity has to be included.

For flow over a flat plate, the boundary layer's thickness grows roughly like the square root of distance along the plate,

$$\delta \sim \sqrt{\frac{\nu x}{U}},$$

where  $\nu = \mu/\rho$ . The square-root growth is a signature of a diffusion-like process: friction at the wall gradually "recruits" more and more fluid to be slowed down as you move downstream, the same way heat spreads outward from a hot object, with the affected region growing as the square root of elapsed time (here, downstream position stands in for elapsed time, since a fluid particle takes longer to reach farther-downstream points).

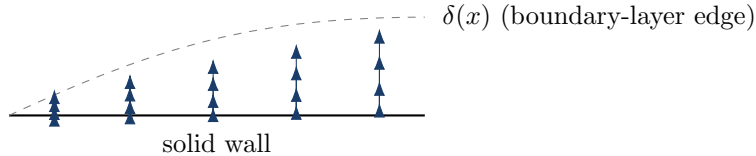


Figure 3: Velocity profile in a growing boundary layer: zero speed at the wall (no-slip), rising to the free-stream value at the edge  $\delta(x)$ , which thickens downstream.

## 12 Adverse Pressure and Flow Separation

When pressure rises in the direction of flow (called an adverse pressure gradient), it's like trying to run up a hill: the flow has to fight against it. Fluid out in the free stream has plenty of speed (and therefore kinetic energy) in reserve and can usually push through. But fluid right next to the wall, inside the boundary layer, has already lost a lot of its speed to friction, so it doesn't have much energy left. If the pressure rise is strong enough, that near-wall fluid can actually stall out and get pushed backward.

The point where the flow right at the wall reverses direction is called separation. Once it happens, the boundary layer peels away from the surface entirely, and a large, swirling, low-pressure wake forms behind the object. This is what happens on a wing at too steep an angle of attack (stall), and it's why round, blunt shapes like spheres and cylinders create large wakes and a lot of drag.

## 13 Vortex Shedding

Behind a blunt object like a cylinder, the flow separates on both the top and bottom, forming two shear layers. These shear layers are unstable, and any tiny imbalance gets amplified: first one side rolls up and sheds a swirling vortex, which changes the pressure just enough to trigger shedding from the other side next, and so on, back and forth, in a steady rhythm. This alternating pattern is called a von Kármán vortex street, and its frequency is described by the Strouhal number,

$$St = \frac{fD}{U}.$$

For a given shape,  $St$  stays remarkably constant across a wide range of speeds, which is a handy rule of thumb for estimating shedding frequency. The same back-and-forth mechanism shows up in a flapping flag and in wing buffet.

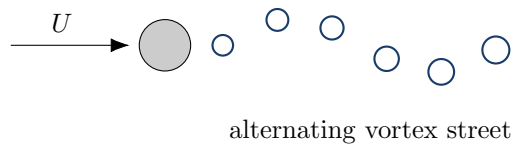


Figure 4: Vortices shed alternately from top and bottom of a bluff body, creating an oscillating side force.

## 14 From Instability to Turbulence

Vortex shedding is already a kind of instability, but a fairly orderly one — a single dominant rhythm. Push the Reynolds number even higher and the shear layers become unstable in many different ways at once, at many different sizes simultaneously, and these disturbances start interacting with each other. The flow loses any simple rhythm and becomes turbulent: chaotic-looking, three-dimensional, and essentially impossible to predict in fine detail, even though it's still governed by ordinary physical laws.

Rather than trying to track every chaotic wiggle, it's common to split the velocity into a steady average part and a fluctuating part on top of it,

$$u = U + u',$$

where  $U$  is the time-averaged speed and  $u'$  is how much the instantaneous speed wiggles around that average. Turbulence contains motion over a huge range of sizes at once, from large swirls down to tiny ones, which brings us to the energy cascade.

## 15 The Turbulent Energy Cascade

Energy usually enters a turbulent flow at large scales — set by the size of the wake or the object producing the turbulence. Those large swirls are themselves unstable and break apart into somewhat smaller swirls, which break apart into smaller ones still, passing energy down the chain with very little lost at each step. Eventually the swirls become so small that friction (viscosity) finally becomes effective at converting their motion into heat, and the cascade stops. This picture — big swirls feeding smaller swirls, all the way down to where friction takes over — is often summarized by Richardson's rhyme: *"big whirls have little whirls that feed on their velocity, and little whirls have lesser whirls, and so on to viscosity."*

The size of the smallest swirls (the Kolmogorov scale) shrinks very quickly as the Reynolds number increases. That means a real, high-speed aerodynamic flow contains swirls ranging from meters across down to a fraction of a millimetre, all active at the same time. Since a computer simulation would need a mesh point for every one of those tiny swirls to capture them directly, the cost of simulating turbulence exactly explodes at high Reynolds number — this is the central reason turbulence is hard to simulate.

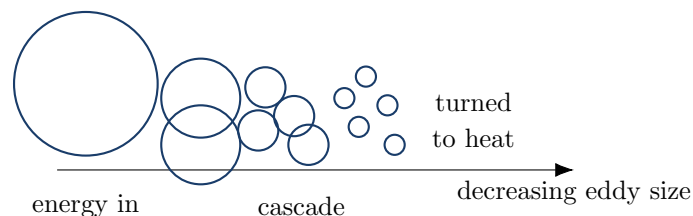


Figure 5: The turbulent energy cascade: large swirls break into smaller ones, passing energy down until friction finally converts it to heat.

## 16 Why We Need CFD

We know the physical laws governing air exactly, but for any realistic shape — a wing, a car, a building — the equations are simply too complicated to solve with pencil and paper. That’s true even before turbulence gets involved, and turbulence, with its huge range of swirl sizes, makes things much worse. Analytical (pencil-and-paper) solutions only exist for a handful of idealized, simplified cases, like the frictionless flow around a circle or an ideal airfoil shape.

Computational Fluid Dynamics (CFD) doesn’t change the underlying physics; it just trades an equation we can’t solve exactly for a similar, approximate problem that a computer can solve numerically — much like replacing a smooth curve with a series of short straight line segments when you don’t have a formula for the curve itself.

## 17 What a CFD Mesh Actually Does

A CFD simulation chops the region of interest into a large number of small cells, forming a mesh. Instead of knowing the exact speed at every single point in space (infinitely many points), the simulation only keeps track of an approximate value in each cell.

This has an important consequence: anything smaller than a single cell — a small swirl, a thin boundary layer — simply cannot be seen by the simulation, the same way a feature smaller than one pixel can’t appear in a digital photo. That’s why mesh resolution matters so much near walls and in wakes, and why simply making the mesh finer *everywhere* is usually far too expensive: it’s much smarter to put the fine resolution only where the flow actually needs it.

## 18 The Basic Idea Behind Finite-Volume CFD

Picture the mesh as a huge collection of small boxes packed together to fill the domain. For each box, the simulation keeps a simple ledger: however much mass, momentum, or energy is inside the box can only change because of flow crossing in or out through its faces (plus any local sources). This is exactly the same “what goes in must come out” bookkeeping from Section 2, just applied to every tiny box separately.

Because whatever one box computes as leaving through a shared face is exactly what its neighbour receives as entering through that same face, nothing ever gets silently created or lost at the boundaries between cells — even on a coarse, irregular mesh, the total amount of mass, momentum, and energy is tracked exactly. Writing this ledger for every box at once, and solving all of them together, is what a CFD solver actually does under the hood, and it boils down to a very large system of linear equations that a computer can grind through.

## 19 How a Computer Approximates a Slope

A true derivative is defined using an infinitely small step size, but a computer only ever has values at a finite set of points, spaced some distance  $\Delta x$  apart. The simplest way to estimate a slope is to use two nearby points and compute rise over run:

$$\left. \frac{du}{dx} \right|_i \approx \frac{u_{i+1} - u_{i-1}}{2\Delta x}.$$

This is just the slope of the straight line connecting two neighbouring points, used as a stand-in for the true slope of the underlying curve. The smaller you make  $\Delta x$ , the better this approximation gets, which is why refining the mesh generally makes a CFD simulation more accurate.

## 20 Convergence Is Not the Same as Being Right

A CFD solution "converging" just means the computer's iterative solver settled on a stable, self-consistent numerical answer for the particular set of discrete equations it was given. That says nothing about whether those equations, on that particular mesh, with those particular assumptions, actually describe the real flow.

There are several separate things that can go wrong, on top of simple numerical convergence: the mesh might be too coarse, the time step might be too large, the boundary conditions might not match reality, or the turbulence model might be a poor fit for this particular flow. A simulation can converge beautifully and still be answering the wrong question.

A converged numerical solution is not necessarily a correct physical solution.

## 21 Why Turbulence Needs a Model

If you average the equations of motion over time to get rid of the chaotic wiggling of turbulence, something inconvenient happens: extra unknown terms pop out that weren't there before. These represent an additional mixing effect, caused by turbulent swirls carrying fast-moving fluid into slow regions and vice versa, over and above the ordinary friction already in the equations. The trouble is that this extra mixing effect isn't expressed in terms of the average flow we're trying to solve for — it's a genuinely new, unknown quantity. This is called the turbulence closure problem, and it's the reason turbulence modelling exists at all: something extra has to be supplied to close the loop.

## 22 Three Levels of Turbulence Simulation: DNS, LES, and RANS

These three approaches differ in how much of the turbulent cascade gets computed directly, versus estimated with a model.

**DNS (Direct Numerical Simulation)** resolves every swirl, all the way down to the smallest ones, with no modelling shortcuts at all. It's extremely accurate but extremely expensive, since (as we saw in the cascade section) the smallest swirls get rapidly smaller as Reynolds number increases. DNS is mostly a research tool, used on simple shapes at modest speeds.

**LES (Large-Eddy Simulation)** resolves the big, geometry-specific swirls directly, and only estimates the effect of the small ones with a simple model. This is a sensible compromise: the big swirls carry most of the energy and depend a lot on the specific shape, while the small swirls behave in a fairly universal way regardless of shape, so a simple model for them doesn't lose much accuracy.

**RANS (Reynolds-Averaged Navier–Stokes)** doesn't resolve any individual swirl at all — it only computes the time-averaged flow, with all of turbulence's effects folded into a model. This is by far the cheapest option and is used constantly in everyday engineering, but it's only as good as the model backing it up.

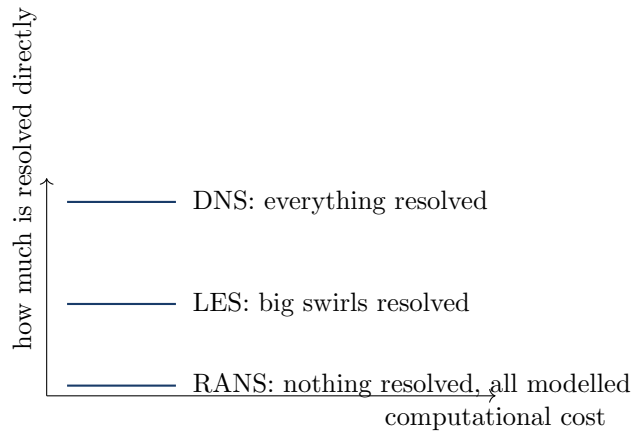


Figure 6: RANS, LES, and DNS trade computational cost against how much turbulence is computed directly rather than modelled.

## 23 The Basic Idea of Eddy-Viscosity Models

The most common turbulence model treats the extra mixing caused by turbulence as if it behaved like an extra, much larger "effective" viscosity, on top of the fluid's real molecular viscosity. This isn't literally true — turbulent swirls are not tiny molecules — but it's a convenient and often reasonably accurate stand-in: wherever the model says the effective (or "eddy") viscosity is large, it means turbulence is mixing things especially vigorously there.

Unlike the real viscosity of a fluid, which is a fixed material property, this effective viscosity changes from place to place depending on how intense the local turbulence is, so the model needs extra equations of its own just to estimate it at each point in the flow. That's what the specific RANS models in the next section actually do.

## 24 Common RANS Models

The most widely used models come in "two-equation" varieties, meaning they solve two extra equations (on top of the main flow equations) to estimate the effective viscosity everywhere.

The  $k-\epsilon$  model tends to work well out in the open flow, away from walls, but needs special extra care right next to walls. The  $k-\omega$  model is naturally better behaved close to walls, which matters for predicting friction and the onset of separation, but is more sensitive to conditions specified far away in the free stream. The SST  $k-\omega$  model is a practical blend of the two: it behaves like  $k-\omega$  near walls and like  $k-\epsilon$  farther out, which is why it's become a common default choice for aircraft and wing simulations.

None of these models is derived from first principles the way Newton's laws are — they're tuned against experiments and more detailed simulations for a range of test cases, and they can give poor answers when pushed outside the conditions they were tuned for. Which model to use should depend on the flow you're actually trying to simulate.

## 25 Checking Your Work: Verification and Validation

**Verification** asks: did we solve the equations correctly? This is checked by making the mesh finer and finer and seeing whether the answer stops changing, checking that the solver’s residuals have dropped to a very small value, and confirming quantities like mass are being conserved exactly. This is a purely numerical, mathematical question.

**Validation** asks: are these even the right equations and models to describe the real flow? This can only be checked by comparing against real experimental data or other trusted results — it’s a physical question, not a mathematical one.

A simulation can be perfectly verified (mesh-independent, tightly converged) and still be wrong, if the turbulence model or boundary conditions don’t match reality. Good practice always checks both, verification first.

## 26 Using CFD for Design, Not Just Prediction

So far we’ve been asking: given a shape, what flow results? Design flips that question around: given a flow behaviour you want (low drag, a particular pressure pattern), what shape produces it?

This ”inverse” question is much harder than the forward one, because many different shapes might come close to the goal, or none might hit it exactly. In practice, engineers define a single number that captures ”how good” a candidate shape is (for example, its drag), and then search systematically for the shape that makes that number as small as possible, often letting a computer nudge the geometry a little at a time, re-run the CFD simulation, check whether the number improved, and repeat. This is the basic idea behind aerodynamic shape optimization, and it’s the same logic — computer-guided trial-and-improvement, informed by CFD — used in far more advanced design methods you’ll encounter in later years.

## 27 Measuring How Much a Flow Wiggles

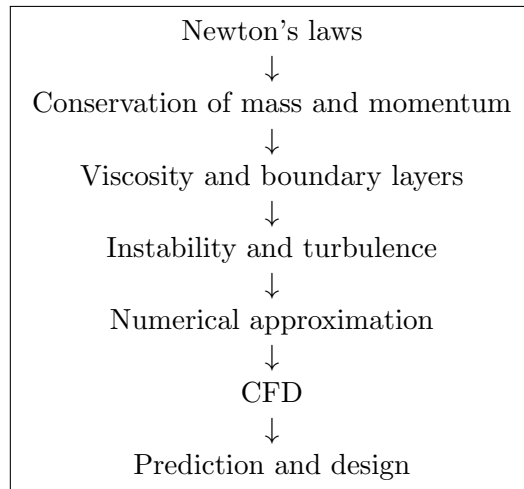
If a signal fluctuates around some average value, simply averaging the wiggle itself gives zero, since positive and negative excursions cancel out. A better way to measure ”how much” something is fluctuating is to square the wiggle first (which removes the sign), average that, and then take the square root to get back to sensible units:

$$u_{\text{rms}} = \sqrt{\langle (u - U)^2 \rangle}.$$

This single number, called the root-mean-square (rms) fluctuation, is a handy way to compare how ”jittery” or unsteady different flows are — useful, for example, in comparing how much buffeting different wing designs experience.

## 28 Putting It All Together

The overall story of this handout is:



CFD is powerful because it combines exact physical laws with approximate numerical methods to tackle problems no one can solve by hand. But that same power is exactly why it's dangerous to trust blindly: a computer can produce a smooth, confident-looking, perfectly converged answer that is still physically wrong.

**The fundamental question is not “Can we simulate it?”  
It is “Do we understand what the simulation is actually telling us?”**